Answer key and rationales

CV-7 Conduction System and the Electrocardiogram

Cardiovascular System (CARD) · NPLEX Part I · NPLEX-P1-CARD-07

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-07-01-C01-BD1

Brain dump 1 · foundational retrieval: Draw the conduction pathway

A competent response contains

· Order: SA node, atrial muscle with the anterior interatrial band to the left atrium, AV node, bundle of His, right and left bundle branches with anterior and posterior left fascicles, subendocardial Purkinje fibers, ventricular muscle.

· SA node lies under the epicardium of the right atrium at the junction with the superior vena cava, at the upper end of the sulcus terminalis.

· AV node lies in the lower interatrial septum in the triangle of Koch, bounded by the coronary sinus opening, the septal tricuspid leaflet, and the tendon of Todaro.

· The His bundle pierces the fibrous skeleton and runs along the lower border of the membranous septum.

· Velocities: SA node about 0.05 m/s, atrial muscle about 0.3 m/s, AV node about 0.05 m/s and slowest, Purkinje fibers about 1.5 to 4 m/s and fastest, ventricular muscle about 0.3 to 0.5 m/s.

· Intrinsic rates: SA node about 100 bpm denervated and 60 to 100 bpm at rest with vagal tone, AV junction about 40 to 60 bpm, Purkinje about 20 to 40 bpm, and atrial and ventricular muscle have none normally.

· Upstroke: L-type Ca2+ current in the SA and AV nodes, fast Na+ current in atrial muscle, Purkinje fibers, and ventricular muscle.

· Conduction slows in the AV node because its cells are small, poorly coupled by few gap junctions, rest near -60 mV where fast Na+ channels are inactivated, and depolarize through slow L-type Ca2+ channels.

· The fibrous skeleton is an electrical insulator, so the AV node and His bundle are the only normal route from atria to ventricles.

· Purkinje fibers activate apex and septum before the base and drive the wall from endocardium to epicardium; the right bundle runs through the moderator band to the anterior papillary muscle so it tenses its chordae first.

Distinctions that matter

· Conduction velocity and intrinsic rate are separate properties: Purkinje fibers conduct fastest but pace slowest.

· Gap junctions couple all myocytes, while the conduction system imposes the order and timing on that coupling.

Common misconceptions

· Students place the AV node in the AV septum generally; it sits in the lower interatrial septum within the triangle of Koch.

· Writing that nodal cells use fast Na+ channels misses why they are slow; their upstroke is L-type Ca2+ and their resting voltage already inactivates most Na+ channels.

Where it is taught: Section C01 of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C01-BD2

Brain dump 2 · integration and application: What the AV delay buys and what happens without it

A competent response contains

· AV delay totals about 0.13 s: about 0.09 s in the node and about 0.04 s in the penetrating bundle, reached about 0.03 s after the impulse leaves the SA node.

· The delay lets atrial systole finish before the AV valves close, adding up to about 20% of resting filling, raising end-diastolic volume and stroke volume through the Frank-Starling mechanism.

· That contribution matters more when the ventricle is stiff and when the rate is fast, because filling time is short.

· At an atrial rate of 400 per minute the node blocks most impulses because its refractory period outlasts its action potential, so only impulses arriving at a recovered node conduct.

· The ventricular response is therefore slower than the atrial rate and irregularly irregular, with a narrow QRS because the His-Purkinje system is still doing the distributing.

· An accessory pathway is fast Na+ dependent muscle without decremental conduction, so it removes both the delay and the filter and passes far more impulses to the ventricle.

· Myocyte to myocyte spread, as in bundle branch block or a ventricular ectopic beat, makes contraction less synchronous.

Distinctions that matter

· Decremental conduction is progressive slowing with faster impulse arrival; refractoriness outlasting the action potential is what actually blocks impulses in flutter and fibrillation. Both come from time-dependent Ca2+ channel recovery.

· Atrial kick raises filling volume; it does not change the rate at which the ventricle is activated.

Common misconceptions

· Students say the atria push in most of ventricular filling. Most filling is passive, and the atrial contribution is up to about 20% at rest.

· Describing the ventricular rate in atrial fibrillation as regular misses that conduction depends on when each impulse finds the node recovered.

Where it is taught: Section C01, final paragraph of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C02-BD1

Brain dump 1 · foundational retrieval: Draw the SA nodal action potential

A competent response contains

· Maximum diastolic potential is about -60 mV and the upstroke begins at about -40 mV.

· Phase 4: HCN channels open on repolarization and carry the funny current If, mainly Na+ inward; delayed rectifier K+ current is switching off; late in phase 4, T-type Ca2+ current and sarcoplasmic reticulum Ca2+ release extruded by the Na+/Ca2+ exchanger add inward current.

· Phase 0: L-type Ca2+ channels open at about -40 mV, giving a slow upstroke of small amplitude.

· Phase 3: L-type channels inactivate and delayed rectifier K+ efflux returns the cell to about -60 mV.

· Nodal cells have very few inward rectifier K+ channels (IK1), so nothing clamps them near the K+ equilibrium potential and the membrane drifts to threshold after every beat.

· Working myocytes rest near -85 to -90 mV because IK1 holds them there.

· The nodal tracing has no phase 1 notch and no phase 2 plateau; refractoriness still outlasts repolarization because Ca2+ channel recovery is time dependent.

Distinctions that matter

· If is an inward Na+ current through HCN channels, not a K+ current, and it switches on with hyperpolarization rather than depolarization.

· T-type Ca2+ current belongs to late phase 4; L-type Ca2+ current carries phase 0.

Common misconceptions

· Calling -60 mV a resting potential: it is the most negative point of a continuously drifting membrane, so the term is maximum diastolic potential.

· Drawing a fast tall upstroke for the SA node. The Ca2+ upstroke is slow and small, which is why nodal conduction velocity is about 0.05 m/s.

Where it is taught: Section C02, opening paragraph and bullets of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C02-BD2

Brain dump 2 · integration and application: Vagal and sympathetic drive at the node

A competent response contains

· Rate is set by the slope of phase 4, the maximum diastolic potential, and the threshold.

· Acetylcholine at M2 receptors acts through Gi: the alpha subunit lowers cAMP and reduces If, flattening the slope; the beta-gamma subunits open IKACh so the maximum diastolic potential becomes more negative; reduced Ca2+ current raises the effective threshold. The cell starts lower, climbs more slowly, and has further to climb.

· Norepinephrine at beta-1 receptors acts through Gs: cAMP rises and binds HCN channels directly to increase If, and protein kinase A phosphorylates L-type Ca2+ channels, so phase 4 steepens and threshold is reached sooner.

· In the AV node, vagal activity slows conduction and lengthens the PR interval, and sympathetic activity speeds conduction and shortens it.

· Strong vagal discharge, as in the baroreceptor reflex or a vasovagal episode, can produce a sinus pause or transient AV block.

· An L-type Ca2+ channel blocker slows sinus rate and AV conduction and reduces contractility, with little effect on the ventricular Na+ upstroke, because nodal phase 0 and myocyte contraction both depend on Ca2+ entry.

· A selective If blocker slows the rate alone, because HCN channels are confined to the pacemaker phase 4 current and play no part in contraction.

Distinctions that matter

· Vagal slowing works on three variables at once; sympathetic acceleration works mainly on the slope of phase 4.

· Both an If blocker and a Ca2+ channel blocker slow the sinus rate, but only the Ca2+ blocker changes AV conduction and contractility.

Common misconceptions

· Saying acetylcholine hyperpolarizes nodal cells by reducing If. The hyperpolarization comes from IKACh opening; reduced If flattens the slope.

· Treating cAMP as working only through protein kinase A. cAMP binds HCN channels directly and shifts their activation to less negative voltages.

Where it is taught: Section C02, paragraphs on rate and autonomic control, and the stop and predict box of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C02-RT01

Retrieval exercise: Rebuild the sympathetic acceleration sequence

A competent response contains

· Norepinephrine binds beta-1 receptors on nodal cells.

· Gs activates adenylyl cyclase and cAMP rises.

· cAMP binds HCN channels directly and If increases; the binding shifts HCN activation to less negative voltages, so the channels open earlier in repolarization.

· Protein kinase A phosphorylates L-type Ca2+ channels, increasing Ca2+ current and making the upstroke reach threshold more readily.

· Phase 4 steepens and threshold is reached sooner.

· Rate rises and AV conduction speeds, which shortens the PR interval.

· The redrawn slope is steeper and reaches the same threshold in less time, from the same maximum diastolic potential.

Distinctions that matter

· cAMP acts on HCN channels without protein kinase A; protein kinase A is the route to the L-type Ca2+ channel.

· Sympathetic drive changes the slope, not the threshold voltage itself.

Common misconceptions

· Drawing a more negative maximum diastolic potential with sympathetic drive. That is the vagal effect through IKACh.

Where it is taught: Section C02, flow diagram on sympathetic acceleration of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C03-BD1

Brain dump 1 · foundational retrieval: Blood supply of the conduction system

A competent response contains

· SA nodal artery: proximal right coronary artery in about 60% of hearts, circumflex in about 40%.

· AV nodal artery: from the artery crossing the crux near the origin of the posterior descending artery, so the right coronary artery in about 80% (right dominance).

· Bundle of His: dual supply from the AV nodal artery and the first septal perforator of the left anterior descending artery.

· Right bundle branch: septal perforators of the left anterior descending artery, which supply the anterior two thirds of the septum.

· Left anterior fascicle: septal perforators of the left anterior descending artery.

· Left posterior fascicle: left anterior descending septal branches and the posterior descending artery, and it is broad, which is why isolated posterior fascicular block is uncommon.

· The His bundle and the left posterior fascicle each have more than one source, so a single artery occlusion is less likely to silence them.

Distinctions that matter

· The SA nodal artery and the AV nodal artery have different percentages and different anatomical origins, so keep the two figures apart.

· Above the node the supply is nodal and usually right coronary; below the node it is septal and usually left anterior descending.

Common misconceptions

· Assuming the right coronary artery always feeds both nodes. It feeds the SA node in about 60% and the AV node in about 80%, and dominance decides the AV figure.

Where it is taught: Section C03, first paragraph of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C03-BD2

Brain dump 2 · integration and application: Why inferior and anterior infarcts block differently

A competent response contains

· Inferior infarct with ST elevation in II, III, and aVF: right coronary artery occlusion, SA and AV nodal tissue at risk, block at the AV node, junctional escape that is narrow at about 40 to 60 bpm, usually transient.

· Anterior infarct with ST elevation in V1 to V4: left anterior descending occlusion, loss of the right bundle and left anterior fascicle, block at the His bundle or bundle branches, ventricular escape that is wide at about 20 to 40 bpm, often persistent with a worse prognosis.

· Non-necrotic mechanisms in the inferior infarct: ischemia depresses the Ca2+ dependent nodal action potential, adenosine released from ischemic tissue acts on A1 receptors to open the same K+ channels as acetylcholine, and vagal afferents in the inferoposterior wall drive reflex bradycardia and hypotension.

· That reflex is the Bezold-Jarisch reflex.

· Nodal tissue usually survives and part of the cause is reflex and metabolic, which is why inferior block recovers.

· In a left dominant heart the circumflex supplies the AV nodal artery, so a circumflex occlusion can cause AV nodal block; because the circumflex feeds the SA node in about 40%, a lateral infarct can cause sinus node dysfunction.

Distinctions that matter

· Level of block decides escape rate and QRS width: nodal block leaves a pacemaker above the bundle branches, so the escape is narrow and faster.

· Transient against persistent follows mechanism: reflex and metabolic depression recovers, necrosis does not.

Common misconceptions

· Reading any complete heart block after infarction as the same problem. The artery sets the level, and the level sets the escape and the prognosis.

· Attributing inferior infarct bradycardia only to nodal ischemia and leaving out adenosine and the vagal reflex.

Where it is taught: Section C03, paragraphs on inferior and anterior infarction and the comparison table of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C04-BD1

Brain dump 1 · foundational retrieval: Draw and label one normal beat

A competent response contains

· P wave is atrial depolarization, under 0.12 s and under 2.5 mm in lead II.

· PR interval runs from the onset of P to the onset of QRS and covers atrial, AV nodal, and His-Purkinje conduction, normally 0.12 to 0.20 s.

· The PR segment is flat because the AV node and His bundle are too small to register, so their delay shows only as elapsed time.

· QRS is ventricular depolarization, 0.06 to 0.10 s, with 0.12 s or more abnormal.

· ST segment corresponds to the plateau, phase 2, and is isoelectric because no voltage gradient exists while the whole ventricle sits at the plateau.

· T wave is repolarization, phase 3, in the same direction as the QRS and inverted in aVR.

· QT interval covers depolarization plus repolarization and approximates ventricular action potential duration; QTc is under about 0.45 s in men and 0.46 s in women.

· U wave is late repolarization, small normally and prominent in hypokalemia.

· Atrial repolarization is hidden within the QRS.

· The T wave is upright where the QRS is upright because epicardial action potentials are shorter, so repolarization runs epicardium to endocardium, the reverse of depolarization.

· At 25 mm/s a small box is 0.04 s and a large box is 0.20 s; 10 mm equals 1 mV.

Distinctions that matter

· PR interval includes the P wave; the PR segment is only the flat part after it.

· The ECG records electrical events, not contraction, and the nodes are inferred from timing rather than seen.

Common misconceptions

· Explaining the upright T wave by saying repolarization travels the same way as depolarization. It travels the opposite way, and the opposite sign of repolarization makes the deflection upright again.

· Calling an isoelectric ST segment a sign of electrical silence. The ventricle is depolarized throughout the plateau, with no gradient to record.

Where it is taught: Section C04, paragraphs on the waveform and the table of normal values of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C04-BD2

Brain dump 2 · integration and application: Work a strip from the numbers

A competent response contains

· Rate is 300 divided by 3 large boxes, which is 100 bpm.

· PR is 6 small boxes times 0.04 s, which is 0.24 s, above the normal 0.12 to 0.20 s, so this is first degree AV block.

· QRS is 3 small boxes times 0.04 s, which is 0.12 s, at the upper limit and abnormal, so ventricular activation is not purely through the His-Purkinje system.

· QT is 8 small boxes times 0.04 s, which is 0.32 s.

· RR is 3 large boxes times 0.20 s, which is 0.60 s; the square root of 0.60 is about 0.77, so QTc is about 0.32 divided by 0.77, or about 0.42 s, which is normal.

· Axis: lead I positive with aVF negative places the axis between 0 and -90 degrees, and a positive lead II keeps it between 0 and -30 degrees, so the axis is normal.

· Rhythm: an upright P in leads I and II before every QRS at a rate of 100 bpm is sinus rhythm, with first degree AV block and a prolonged QRS alongside it.

Distinctions that matter

· A positive lead II with I positive and aVF negative means a normal axis; a negative lead II in the same situation means left axis deviation.

· Rate normality and rhythm are separate judgments from interval normality: the rhythm here is sinus even though the PR and QRS are abnormal.

Common misconceptions

· Reading an abnormal PR as a rhythm diagnosis. Every P conducts here, so this is first degree block within sinus rhythm.

· Comparing the raw QT with the normal limit. QT must be corrected for rate before judging it.

Where it is taught: Section C04, paragraphs on rate, intervals, and axis, and the worked example of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C05-BD1

Brain dump 1 · foundational retrieval: Build the AV block table from memory

A competent response contains

· First degree: PR constant and over 0.20 s with every P conducted, site the AV node, QRS narrow, low risk of progression.

· Mobitz I: PR lengthens until a P fails to conduct and then resets, site the AV node, QRS narrow, low risk.

· Mobitz II: PR of conducted beats constant with sudden dropped P waves, site the His bundle or bundle branches, QRS often wide, high risk of progression to complete block.

· Third degree: PR varies because the atria and ventricles are dissociated, site either nodal or infranodal, QRS narrow if the escape is junctional and wide if it is ventricular, outlook depends on the escape.

· In Mobitz I the PR increment is largest in the beat after the pause, so RR intervals shorten before the dropped beat and the complexes form groups.

· In third degree block the P waves and QRS complexes march independently, which is AV dissociation.

· Causes: fibrosis, myocardial infarction, Lyme carditis from Borrelia burgdorferi, sarcoidosis, and maternal anti-Ro/SSA antibodies injuring the fetal AV node.

· First degree block also arises from athletic vagal tone, nodal-slowing drugs, and degeneration.

Distinctions that matter

· Gradual PR change locates the block in the AV node; all-or-none failure locates it in His-Purkinje tissue.

· Mobitz II and third degree are not distinguished by QRS width alone: in Mobitz II some P waves still conduct with a fixed PR, while in third degree none conduct.

Common misconceptions

· Expecting RR intervals to lengthen before the dropped beat in Mobitz I. They shorten, because each PR increment is smaller than the one before.

· Treating first degree block as a dropped beat. Every P conducts, just slowly.

Where it is taught: Section C05, paragraphs on the degrees of block and the AV block table of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C05-BD2

Brain dump 2 · integration and application: Gradual failure against all-or-none failure

A competent response contains

· AV nodal cells depolarize through slow L-type Ca2+ channels whose recovery is time dependent; His-Purkinje cells use fast Na+ channels.

· Time-dependent recovery means the node conducts each closely spaced impulse more slowly, which is decremental conduction, so failure is gradual.

· His-Purkinje tissue conducts at nearly full speed or not at all, so infranodal block appears suddenly.

· Mobitz I: each successive impulse arrives at a less recovered node, so the PR lengthens until one impulse is blocked, after which the node recovers and the cycle restarts.

· Mobitz II: diseased His or bundle branch tissue fails without warning, so conducted beats keep a constant PR and P waves drop abruptly.

· Escape after nodal block is junctional, narrow, about 40 to 60 bpm, and uses the normal bundle branches; escape after infranodal block is a distal Purkinje or ventricular pacemaker, wide, about 20 to 40 bpm, spreading myocyte to myocyte, and liable to fail.

· In 2:1 block the PR cannot be compared from beat to beat because only every other P conducts, so no trend is visible; a wide QRS favors an infranodal site.

· When the last fascicle of a bifascicular block fails, the result is complete infranodal block with a wide escape at about 20 to 40 bpm, slow enough to cause syncope.

Distinctions that matter

· Decremental conduction is a normal property of the AV node that becomes visible as Wenckebach when the node is stressed or diseased; it is not a property of His-Purkinje tissue.

· QRS width reports where ventricular activation starts, which is why it helps locate a block the PR cannot.

Common misconceptions

· Assuming 2:1 block with a narrow QRS must be nodal. A narrow QRS makes a nodal site more likely but does not settle it from the surface tracing.

· Describing the junctional escape as unreliable. It is the faster and more dependable of the two because it still uses the bundle branches.

Where it is taught: Section C05, opening paragraph, block descriptions, and the final paragraph on fascicular block of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C05-CP01

Comparison table: Right against left bundle branch block

A competent response contains

· Septal depolarization: RBBB keeps the normal left to right direction because the left bundle is intact; LBBB reverses it to right to left.

· V1: RBBB shows rSR prime; LBBB shows a deep wide QS or rS.

· I and V6: RBBB shows a wide slurred S wave with the septal q wave kept; LBBB shows a broad notched R wave with the septal q lost.

· Septal q wave in I, aVL, V5, and V6: present in RBBB, absent in LBBB.

· Second heart sound: RBBB gives wide splitting; LBBB gives paradoxical splitting.

· QRS duration: 0.12 s or more in complete block of either type, and about 0.10 to 0.12 s with the same morphology in incomplete block.

· Causes: RBBB from right ventricular strain such as pulmonary embolism or atrial septal defect, and from septal infarction, and it can occur in a normal heart; LBBB almost always means structural disease such as ischemia, hypertension, aortic stenosis, or cardiomyopathy, and it can mask a new infarct.

· In either block the affected ventricle is activated late by myocyte to myocyte spread, the terminal QRS points toward that ventricle, and ST-T changes point away from it.

Distinctions that matter

· The septal q wave is the quickest discriminator: it survives in RBBB and disappears in LBBB, because LBBB reverses the direction of septal activation.

· RBBB can be a normal variant; LBBB implies structural heart disease.

Common misconceptions

· Reading ST-T changes in a bundle branch block as ischemia. They are secondary to the abnormal depolarization and point opposite the terminal QRS.

· Thinking incomplete block has a different pattern. The morphology is the same, only the QRS is 0.10 to 0.12 s.

Where it is taught: Section C05, bundle branch paragraph and the RBBB against LBBB table of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C06-BD1

Brain dump 1 · foundational retrieval: Mechanisms of tachyarrhythmia, and the premature beats

A competent response contains

· Abnormal automaticity: a steep phase 4 in tissue outside the SA node, from catecholamines, stretch, ischemia, or electrolyte disturbance.

· Triggered activity: early afterdepolarizations during a prolonged plateau, or delayed afterdepolarizations in Ca2+ overloaded cells.

· Reentry: the commonest cause of sustained tachycardia, needing two pathways, unidirectional block in one of them, and conduction slow enough that the blocked limb recovers before the impulse returns.

· Wavelength is conduction velocity times refractory period, and the circuit must exceed the wavelength for reentry to persist.

· Slow conduction from ischemia or scar, and short refractoriness from ischemia or atrial remodeling, both shorten the wavelength and make reentry easier to sustain.

· PAC: early abnormal P wave, narrow QRS, and a pause less than compensatory because the premature impulse resets the SA node.

· PVC: early wide QRS with no preceding P wave and a T wave discordant with the QRS, with a fully compensatory pause because retrograde block in the AV node protects the sinus rhythm from being reset.

· A PVC falling on the peak of the T wave, the vulnerable period, can start ventricular fibrillation.

Distinctions that matter

· The pause distinguishes the two premature beats and both pauses follow from what reaches the SA node: a PAC resets it, a PVC does not.

· Early afterdepolarizations arise during repolarization from a long plateau; delayed afterdepolarizations arise after repolarization from Ca2+ overload.

Common misconceptions

· Calling slow conduction harmful only. Reentry needs it, which is why ischemic and scarred tissue is arrhythmogenic.

· Saying a PVC resets the sinus node. The retrograde impulse is blocked in the AV node, so the sinus rhythm continues on schedule.

Where it is taught: Section C06, opening paragraph, reentry cycle, and premature beat paragraph of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C06-BD2

Brain dump 2 · integration and application: Read four strips back to their mechanisms

A competent response contains

· Patient 1: atrial fibrillation from pulmonary vein triggers and a dilated fibrotic atrium sustaining wavelets at about 350 to 600 per minute; the node conducts only impulses that find it recovered, so the ventricular response is far slower than the atrial rate and irregularly irregular with a narrow QRS.

· Patient 1 loses atrial kick from ventricular filling, and stasis in the left atrial appendage promotes embolic stroke.

· Patient 2: atrial flutter, macroreentry around the tricuspid annulus through the cavotricuspid isthmus, with an atrial rate near 300 per minute and 2:1 nodal conduction giving a regular ventricular rate near 150 bpm.

· Patient 3: paroxysmal supraventricular tachycardia, usually AV nodal reentry through dual nodal pathways, regular at 150 to 250 bpm with a narrow QRS and the retrograde P wave buried in or just after the QRS, sometimes as a pseudo R prime in V1; it starts and stops abruptly and vagal slowing of the node can terminate it.

· Patient 4: ventricular tachycardia, three or more consecutive ventricular beats over 100 bpm, usually scar reentry, with a wide QRS and AV dissociation.

· The other two confirming findings in patient 4 are capture beats and fusion beats.

· Narrow QRS means the ventricles were activated through the His-Purkinje system; wide QRS means ventricular origin or aberrant conduction.

Distinctions that matter

· Flutter with 2:1 conduction is regular near 150 bpm, while fibrillation is irregularly irregular; the atrial rate and the filtering node explain both.

· AV nodal reentry and ventricular tachycardia are both regular, and QRS width plus AV dissociation separate them.

Common misconceptions

· Reading a ventricular rate of 130 in atrial fibrillation as the atrial rate. The atria are firing several hundred times per minute and the node blocks most impulses.

· Expecting a visible P wave in AV nodal reentry. The atria and ventricles activate almost together, so it is hidden in or just after the QRS.

Where it is taught: Section C06, atrial arrhythmia paragraph, worked example on nodal reentry, and the tachyarrhythmia table of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C06-RT01

Retrieval exercise: Rebuild the reentrant circuit

A competent response contains

· Step 1: a premature impulse reaches a branch point.

· Step 2: one limb is still refractory, giving unidirectional block.

· Step 3: the impulse travels slowly down the other limb.

· Step 4: the blocked limb recovers during that slow passage.

· Step 5: the impulse returns up the recovered limb.

· Step 6: it re-enters the first limb and the loop repeats.

· Blocking one limb, or lengthening refractoriness, stops the circuit.

· In AV nodal reentry the fast pathway has a long refractory period and the slow pathway a short one. A premature atrial beat finds the fast pathway refractory, descends the slow pathway with a long PR, then travels retrograde up the recovered fast pathway while also descending the His bundle, and re-enters the slow pathway at 150 to 250 bpm.

· Because atria and ventricles activate almost together, the retrograde P wave sits inside or just after the narrow QRS, sometimes appearing as a pseudo R prime in V1.

Distinctions that matter

· Unidirectional block is required at the start, and slow conduction is required to keep the circuit alive; one without the other does not sustain reentry.

· The fast pathway conducts quickly but recovers slowly, which is exactly why a premature beat is blocked there first.

Common misconceptions

· Drawing both limbs as conducting normally at the outset. Without block in one limb the impulse meets itself and extinguishes.

· Expecting a long PR during the tachycardia. The long PR belongs to the initiating premature beat going down the slow pathway.

Where it is taught: Section C06, reentry cycle and the worked example on nodal reentry of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C06-PR01

Prediction exercise: Predict the effect of a low plasma potassium

A competent response contains

· IKr falls, because low plasma K+ reduces IKr and macrolides block IKr directly.

· Less repolarizing K+ current lengthens the ventricular action potential, so the QT interval lengthens.

· The U wave becomes prominent, which is characteristic of hypokalemia.

· During the prolonged plateau, L-type Ca2+ channels have time to recover and reopen.

· That inward Ca2+ current produces early afterdepolarizations, which fire premature beats.

· Uneven repolarization across the ventricle supports reentry, so the tracing shows a polymorphic QRS that twists about the baseline on a background of long QT, which is torsades de pointes, and it can degenerate into ventricular fibrillation.

Distinctions that matter

· Low K+ reduces IKr rather than increasing K+ efflux, which is why hypokalemia lengthens repolarization instead of shortening it.

· Early afterdepolarizations start the rhythm; dispersed repolarization and reentry keep it going.

Common misconceptions

· Expecting a low plasma K+ to make the cell fire less because K+ currents are repolarizing. Reduced IKr prolongs the plateau and invites Ca2+ channel reopening.

· Treating a prominent U wave as a normal variant in this setting. It is the expected finding in hypokalemia.

Where it is taught: Section C06, torsades paragraph and the long QT flow diagram, with the U wave in the C04 table of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C07-BD1

Brain dump 1 · foundational retrieval: Preexcitation and the failing sinus node

A competent response contains

· The accessory pathway exists because the fibrous AV rings incompletely separate atrial from ventricular muscle during development, leaving a bundle of Kent made of fast Na+ dependent muscle that usually lacks decremental conduction.

· Sequence: the atria depolarize normally; the impulse crosses the pathway without nodal delay; part of the ventricle activates early so the PR is under 0.12 s; slow cell to cell spread forms the delta wave; the impulse arriving through the AV node activates the rest of the ventricle rapidly; the result is a wide fusion QRS with secondary ST-T changes.

· On the complex, the pathway produced the short PR and the delta wave, and normal His-Purkinje conduction produced the rest of the QRS.

· Preexcitation alone is the WPW pattern; with episodes of tachycardia it is WPW syndrome. Ebstein anomaly is an association.

· Vagal slowing of the AV node enlarges the delta wave and sympathetic drive can hide it, and a concealed pathway conducts only retrogradely, showing no delta wave while still supporting reentry.

· Sick sinus syndrome is intrinsic failure of the SA node, usually from degenerative fibrosis in older adults, and less often from SA nodal artery ischemia, amyloidosis, sarcoidosis, or inherited channel defects.

· Fewer pacemaker cells give inappropriate sinus bradycardia and chronotropic incompetence, a poor rate rise with exercise; sick sinus syndrome often coexists with AV conduction disease and atrial fibrillation.

Distinctions that matter

· The pathway bypasses the AV node but enters ventricular muscle outside the Purkinje network, which is why early activation is slow and inscribes a delta wave.

· The WPW pattern describes the resting tracing; WPW syndrome adds the arrhythmia.

Common misconceptions

· Saying the whole QRS is preexcited. It is a fusion complex, with the pathway and the AV node each activating part of the ventricle.

· Expecting every accessory pathway to show a delta wave. A concealed pathway conducts retrogradely only.

Where it is taught: Section C07, first paragraph, the WPW flow diagram, and the sick sinus paragraph of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C07-BD2

Brain dump 2 · integration and application: The dangerous pathway and the long pause

A competent response contains

· A refractory period near 0.25 s allows about 240 impulses per minute or more, since 60 divided by 0.25 is 240.

· The AV node would not pass that many because its refractory period is long and time dependent, lengthening further as impulses arrive faster.

· The tracing is very fast, irregular, and wide, with a QRS shape that varies beat to beat as the proportion of fusion between pathway and node changes.

· Short diastole and impulses landing in the vulnerable period allow degeneration to ventricular fibrillation.

· Slowing the AV node does not slow this ventricular rate and can shift more conduction onto the pathway.

· Orthodromic reentry descends the AV node and returns up the pathway, so the QRS is narrow and the delta wave disappears; antidromic reentry descends the pathway and gives a wide, fully preexcited QRS.

· A pause that is an exact multiple of the P to P interval means SA exit block: the impulse forms but cannot leave the node.

· A pause that is not a multiple means sinus arrest: the node fails to fire at all.

· The pause follows the tachycardia because the rapid atrial rhythm overdrive suppresses the node; a healthy node recovers within about 1.5 s, while a fibrotic node with few pacemaker cells recovers slowly, so nothing fires until it or an escape pacemaker reaches threshold.

Distinctions that matter

· Orthodromic tachycardia is narrow and antidromic is wide, and the direction of travel through the pathway is what decides this.

· Sinus arrest against SA exit block is settled by arithmetic on the pause, not by its length.

Common misconceptions

· Assuming nodal slowing controls the ventricular rate in every atrial fibrillation. With an accessory pathway the node is not the limiting route.

· Reading a post-tachycardia pause as a separate new problem. It is overdrive suppression unmasking a diseased node, the tachy-brady form of sick sinus syndrome.

Where it is taught: Section C07, accessory pathway paragraph, sick sinus paragraph, tachy-brady paragraph, and the stop and predict box of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C07-CA01

Clinical application: Palpitations in a young man with a wide, irregular tachycardia

A competent response contains

· Recognize: atrial fibrillation conducted to the ventricles through an accessory pathway, identified on the earlier tracing by a PR of 0.10 s with a delta wave, the slurred upstroke, and a QRS of 0.13 s with discordant ST-T changes.

· Explain: the pathway is fast Na+ dependent muscle without decremental conduction and with a short refractory period, so it passes atrial impulses at a rate the AV node would block; QRS width and shape vary because the degree of fusion between pathway conduction and nodal conduction changes from beat to beat.

· Differentiate: atrial fibrillation conducted through the node is irregularly irregular but narrow and slower; ventricular tachycardia is regular and uniform in QRS shape with AV dissociation, capture beats, and fusion beats, while this rhythm is irregular with QRS complexes that change shape.

· Investigate: on the sinus rhythm tracing, measure the PR interval, which is under 0.12 s, look for a delta wave at the QRS onset, measure the QRS, which is 0.12 s or more, and note ST-T changes directed opposite the QRS; the earlier episodes of abrupt regular palpitations fit orthodromic reentry through the same pathway.

· Decide: the dangerous property is the pathway's short, non-decremental refractory period, which permits a very fast ventricular response, and this rhythm can degenerate into ventricular fibrillation.

· Justify: with a pathway refractory period near 0.25 s about 240 impulses per minute or more reach the ventricle, while the AV node's long time-dependent refractoriness would block most of them; slowing the node therefore does not reduce the rate and can shift still more conduction onto the pathway, and short diastole with impulses landing in the vulnerable period is what allows fibrillation.

Distinctions that matter

· Varying QRS shape points to variable fusion through two routes, which separates this from the uniform wide complexes of scar reentry.

· The hemodynamic picture comes from rate and lost filling time, not from loss of the conduction system itself.

Common misconceptions

· Classifying every irregular wide-complex tachycardia as ventricular tachycardia. Preexcited atrial fibrillation is irregular with changing QRS morphology.

· Assuming the AV node still governs the ventricular rate once a pathway is conducting.

Where it is taught: Section C07, accessory pathway paragraphs, with reentry and fibrillation mechanisms in C06 of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C03-Q01

Board-style question: Complete block after an inferior infarction

Correct answer: B. Ischemia of the atrioventricular node with adenosine release and reflex vagal discharge

Why it is right: Right coronary occlusion starves the AV node, and ischemia of its Ca2+ dependent conduction, adenosine acting on A1 receptors to open the same K+ channels as acetylcholine, and the vagal reflex from the inferoposterior wall together produce nodal block with a narrow junctional escape at 40 to 60 bpm.

A is wrong: Necrosis of the His bundle and proximal bundle branches follows anterior infarction from the left anterior descending artery and gives a wide escape at about 20 to 40 bpm, not a 0.08 s QRS at 44/min.

C is wrong: An accessory pathway bypasses the AV node and speeds ventricular activation, producing a short PR with a delta wave rather than bradycardia with dissociated P waves.

D is wrong: Sinoatrial node fibrosis would slow or stop the P waves, but the P waves here are 84/min and normal, and the problem is that they are not conducted.

A competent response contains

· ST elevation in II, III, and aVF marks an inferior infarct, usually from proximal right coronary artery occlusion, which supplies the AV nodal artery in about 80% of hearts.

· Independent P waves at 84/min with QRS complexes at 44/min are complete block with AV dissociation, and the 0.08 s QRS shows the escape pacemaker is junctional, so the block sits in the AV node.

Distinctions that matter

· Escape QRS width locates the block: narrow means the pacemaker is above the bundle branches.

Common misconceptions

· Reading any complete block after infarction as infranodal. The lead group and the narrow escape point to the node.

Where it is taught: Section C03 of the CV-7 Conduction System and the Electrocardiogram notes.

NPLEX-P1-CARD-07-01-C05-Q02

Board-style question: Locating the level of block

Correct answer: D. His bundle and bundle branches

Why it is right: His-Purkinje cells depolarize through fast Na+ channels and conduct at nearly full speed or not at all, so a diseased bundle gives a constant PR with P waves that drop without warning, and the wide rSR prime QRS shows existing bundle branch disease.

A is wrong: Sinoatrial node failure would produce missing P waves and pauses, whereas the P waves here arrive on time and some are simply not conducted.

B is wrong: Atrial muscle disease would alter P wave shape and duration and does not interrupt conduction to the ventricles in this pattern.

C is wrong: Atrioventricular nodal block conducts decrementally, so the PR would lengthen progressively before a beat dropped and the conducted QRS would usually be narrow.

A competent response contains

· A constant PR in every conducted beat with sudden non-conducted P waves is Mobitz II, which is infranodal.

· A QRS of 0.14 s with rSR prime in V1 is right bundle branch block, confirming that the disease lies in the bundle branches.

Distinctions that matter

· Constant PR with abrupt dropped beats is all-or-none His-Purkinje failure, while a lengthening PR before the drop is nodal.

Common misconceptions

· Taking a normal PR interval as evidence that conduction below the node is intact. His-Purkinje tissue conducts at nearly full speed until it fails outright.

Where it is taught: Section C05 of the CV-7 Conduction System and the Electrocardiogram notes.